BHANU PRATAP SINGH RAJPOOT

bhanupratapsinghrajpoot62@gmail.com | bhanupratapraj poot231382@acropolis.in
LinkedIn: Bhanu pratap singh rajpoot | GitHub: github.com/bhanurajpoot04 | Bhounrasa, Dewas, M.P.

CAREER OBJECTIVE

Robotics & R&D Engineer at ROBOTWALA with hands-on experience in robotic, automation and IoT systems on Raspberry Pi 4/5, ESP32 and STM32. Final-year Electronics & Communication student with strong fundamentals in circuit design, sensor interfacing, PCB development and real-time debugging, passionate about turning ideas into real-world robotic and intelligent systems.

EXPERIENCE

R&D Engineer, ROBOTWALA (Indore, on-site) | Jul 2026 – Present
- Build smart automation systems and IoT projects using Raspberry Pi 4/5, ESP32 and embedded programming.
- Design and develop robotic systems and prototypes. Promoted from intern to full-time based on project work.

R&D Intern, ROBOTWALA (Indore, on-site) | Apr 2026 – Jun 2026
- Worked on robotics projects focused on Raspberry Pi integration and precise hardware assembly.

Secretary (Dec 2024 – Dec 2025) and Member (Dec 2025 – Present), IEEE AP-S Student Branch, AITR | Indore
- Organized and coordinated technical events, webinars, seminars and hands-on workshops.

Summer Industrial Training, Acropolis Institute of Technology & Research | Jul – Dec 2025
- Built a toroid winding machine, a smart hydroponics system and a Raspberry Pi sorting conveyor, each with its own PCB.

ACADEMIC RECORD

Bachelor of Engineering (Electronics & Communication)
Acropolis Institute of Technology and Research, Indore (RGPV) | 2023–2027 | **CGPA: 5.55**
Higher Secondary Certificate (10+2)
Saraswati Vidhya Mandir School (MP Board) | 2023 | 67%
Secondary School Certificate (10)
New Silver Bells School (MP Board) | 2021 | 89%

PROJECTS

SCARA Robotic Arm *(Minor Project)*
- Built a 3D-printed 4 DOF arm with stepper joints and a servo gripper for pick-and-place of small objects.
- Wrote the control logic and complete project documentation, including hand-gesture control from a laptop camera in Python.

Weather Monitoring & Control System using ESP *(Completed)*
- Built an IoT weather station that reports temperature and humidity in real time over Wi-Fi.
- Added an on-site LCD display and automated control based on environmental conditions.

Toroid Coil Winding Machine (Arduino Nano) *(Internship Project)*
- Automated winding of copper wire on toroidal cores with a NEMA 17 stepper feeder (A4988) and a 12V DC motor (L293D).
- Counted turns live with an IR sensor on a 16x2 I^2C LCD, with encoder speed and direction control. Designed and fabricated the PCB.

Smart Hydroponics System (ESP32) *(Internship Project)*
- Monitored pH, TDS, temperature and humidity, and scheduled relay-driven 230V and 12V pumps with a DS3231 RTC. Designed the PCB.

Colour & Shape Sorting Conveyor (Raspberry Pi 5) *(Internship Project)*
- Sorted objects on a belt using a Pi Camera with OpenCV colour and contour detection, a servo gate and an L298N motor driver. Designed the PCB.

TRAININGS

- **LLM Training** (Ongoing) – currently training large language models in Python.
- **STM32 Training** – hands-on course with STM32CubeIDE covering GPIO, pin and clock configuration; first project: LED blink with buzzer.

TECHNICAL SKILLS

Hardware & Tools:

- Raspberry Pi 4/5, ESP32, STM32, Arduino
- Sensor interfacing, PWM and motor control
- Basic PCB Design (KiCad – Learning)

Communication Protocols:

- I^2C – sensor and module interfacing
- SPI – full-duplex communication for high-speed peripheral interfacing
- UART – serial communication for debugging and device-to-device data transfer
- Wi-Fi – wireless data transfer for IoT systems

Programming Languages & Tools:

- Python, C++, Embedded C
- OpenCV, STM32CubeIDE, nRF Connect SDK, MATLAB

AREAS OF INTEREST

- Robotics, Embedded Systems & IoT
- Raspberry Pi, Automation and Intelligent Systems
- Research & Development, Hardware Prototyping and Circuit Design
- Sensor Interfacing and Communication Protocols
- PCB Design and Development

CERTIFICATIONS & ACHIEVEMENTS

- nRF Connect SDK Fundamentals – Nordic Semiconductor (DevAcademy), May 2026
- National Level Project Competition (NLPC-26)
- MATLAB Certifications
- Participated in Electronova Hackathon and MSME Hackathon

LEADERSHIP & CO-CURRICULAR ACTIVITIES

- Secretary, IEEE AP-S Student Branch, Acropolis Institute of Technology & Research.
- Volunteered in robotics events at college technical fest Tech-o-tsav.
- Assisted in organizing Robo Soccer Robot competition.
- Volunteered in Robolearn event organized by IEEE-SIGHT.

REFERENCES

Mr. Atul Bharat – Group Director-CDC, Acropolis Group of Institutions
Prof. Udaybhanu Chandravat – HOD-EC, Acropolis Institute of Technology & Research
Contact details available on request.

PERSONAL DETAILS

Gender	: Male
Nationality	: Indian
Languages Known	: Hindi, English, Marwadi
Mother Tongue	: Hindi
Strengths	: Team Collaboration, Problem Solving
Area of Improvement	: Stage fear

DECLARATION

I hereby declare that the information furnished above is true and correct to the best of my knowledge and belief.

Date: 24-10-2026

Signature: _____

Place: Indore

Name: Bhanu Pratap Singh Rajpoot